

11-Gb/s 80-km transmission performance of zero-chirp silicon Mach-Zehnder modulator

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Abstract: The 11-Gbps 80-km transmission performance of a zero-chirp silicon Mach-Zehnder modulator has been characterized. The zero-chirp characteristic of the silicon modulator is confirmed in the constellation measurement, and gives high tolerance both for positive and negative chromatic dispersion. A low-dispersion-penalty transmission up to 80km using the 11-Gbps non return-to-zero on-off-keying format is confirmed via bit-error-rate measurements with a performance comparable to that of a commercial lithium-niobate modulator. The dispersion tolerance at 2-dB power penalty for a bit-error-rate of 10^{-3} is more than ± 950 ps/nm. Further, 22.3-Gbps binary phase-shift-keying is demonstrated, and the back-to-back power penalty with respect to the lithium-niobate modulator is less than 0.5dB.

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1. Introduction

Mach-Zehnder modulators (MZMs) are key components in high-speed optical-fiber telecommunication systems. They provide signal modulation without frequency chirp under push-pull operation [1]. The modulated signals can be transmitted over long distances such as 80 km without serious degradation in a bit-error-rate (BER) even under chromatic dispersion in an optical-fiber link. Furthermore, in phase shift keying (PSK), the zero-chirp operation allows high tolerance to imperfections such as those of the driving voltage and bandwidth [2]. Even in the insufficient conditions, undesirable phase shifts are suppressed in the zero-chirp operation. Lithium niobate (LN) MZMs are extensively used in high-speed long-haul optical-fiber transmission systems.

Silicon modulators have attracted attention because of their potential advantages of reduced foot-print and low fabrication costs using well-developed complementary metal-oxide-semiconductor (CMOS) compatible technologies. Many studies have focused on silicon modulators, and high-speed modulation with transmission rates of more than 40 Gbps via on-off keying (OOK) modulation using such devices has been recently reported [3–5]. However, studies on push-pull operation and long-haul transmission are limited. In the resonance type modulator, some transmission characteristics were reported. Using microring modulator with pre-emphasis driving, 80-km transmission at 12.5 Gbps was successfully demonstrated [6]. Using the microdisk-type modulator, 70-km transmission at 10 Gbps has been achieved [7]. However, for achieving the zero-chirp modulation which provides high tolerance for both positive and negative dispersion, an MZM modulator with a push-pull operation is essential. 125-km transmission at 10.3 Gbps with silicon MZM has been achieved, however, the zero-chirp characteristics including tolerance for negative dispersion have not been discussed [8]. A single-drive push-pull operation using an MZM has been realized and achieved at 50-Gbaud OOK [5] and the similar design has been incorporated in a dual-polarization quadrature PSK (DP-QPSK) modulator [9]. The dispersion of the modulator has been investigated and a remaining frequency chirp has been observed [10].

In this paper, we report the zero-chirp push-pull operation of a silicon MZM in 11.1-Gbps non-return-to-zero (NRZ) OOK, and 22.3-Gbps NRZ- binary phase-shift keying (BPSK). The frequency chirp of the silicon MZM is characterized via constellation measurements and DC characteristics measurements. Using the silicon MZM, we demonstrate 80km single-mode fiber (SMF, G.652) transmission in the 11.1-Gbps NRZ-OOK format with a low dispersion penalty comparable with the performance of LN MZM.

2. Design and experimental setup

The top view and phase-shifter design of the Si MZM are shown in Fig. 1(a). The phase shifter is based on a rib waveguide with the use of a pn junction. The waveguide dimensions are 600-nm rib width, 220-nm rib thickness, and 95-nm slab thickness, which is the same as that reported in [11]. The pn junction is formed at the center of the rib structure. The p-type and the n-type regions are adjacent to each other. Doping concentrations of the p-type and the n-type regions are $5 \times 10^{17} / \text{cm}^3$ and $2 \times 10^{17} / \text{cm}^3$ respectively. The electrode is made of aluminum, and it connects to the PN junction through a contact hole. The contact area of the silicon slab is heavily doped, and it forms the ohmic contact to the electrode. The heavily doped area is 1 μm away from the rib structure. The electrode used is a coplanar traveling-wave electrode, as shown in the figure. The GSGSG-type electrode covers the two arms in the Mach-Zehnder interferometer (MZI), as shown in the top view. Each signal line has a width of 10 μm and the space between ground and signal is a half of the signal width. The Si MZM

consists of the phase shifter with a length of 4 mm, the Y-branch, and an inverted-type mode field converter. The Si MZM is fabricated using 248nm optical lithography on 8-inch SOI wafer with 2μm buried oxide layer thickness. The core of the waveguide is etched from the top silicon layer. The detail of the fabrication process is the same as that reported in [12].

The optical loss in the phase shifter is 3.2 dB, which was extracted from transmittances in different lengths of the phase shifter without applied bias. The fiber-to-fiber loss is less than 10.5 dB in the C and L bands, which was measured as the maximum transmittance in the cyclic spectrum of asymmetric MZI without applied bias. The V_π of the phase shifter is approximately 7V, which is obtained from the measurement of phase shift versus applied bias shown in section 4.

The experimental setup is shown in Fig. 1(b). Both the arms of the Si MZM were driven with RF signals of opposite polarity (DATA, DATA) respectively for push-pull operation. The RF signals were applied to the Si MZM chip through RF probes. Input and output of the Si MZM were coupled with lensed PANDA fibers. The MZI is asymmetric for experimental purpose, which allows the operating point of the MZM to be controlled by adjusting the wavelength.

In the BER measurements, noise was added by an erbium-doped fiber amplifier (EDFA). By adjusting the input power to the EDFA with a variable optical attenuator (VOA), optical signal-to-noise ratio (OSNR) was changed. The OSNR was measured at the input of the bandpass filter with optical spectrum analyzer in a normalized range of 0.1-nm noise bandwidth. The input power to the PD in the receiver was maintained constant during the measurements.

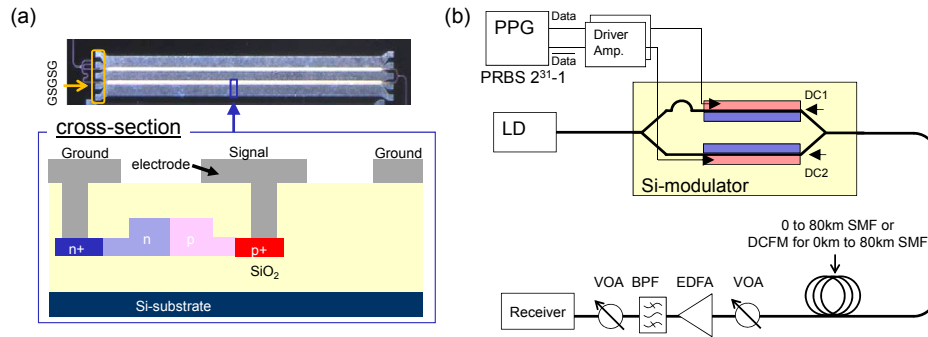


Fig. 1. (a) Top view of the silicon MZM with phase shifter design and (b) experimental setup for push-pull operation and BER measurement

3. Experimental results

Firstly, we present the experimental evidence of zero-chirp operation of the Si MZM in the constellation diagrams in Fig. 2. Phase shift in the bit-transition was acquired using a real-time oscilloscope (Agilent DSAX93204A, which has an electrical bandwidth of 32GHz and a sampling rate of 80GS/s) in the homodyne coherent-detection configuration, where an input LD is also used as the local oscillator. The constellation diagrams in Figs. 2(a) and 2(b) represent the characteristics of the NRZ-OOK at 11.1 Gbps in the Si MZM and the commercial LN MZM, respectively. The driving voltage of the Si MZM was 3.5 V_{pp} for each arm with 4V DC reverse bias, which is full modulation with extinction ratio (ER) of 11 dB with 50% cross-point of the eye-diagram. The LN MZM is a zero-chirp modulator with 13GHz bandwidth. The driving voltage of the LN is about 4 V_{pp} with full modulation in OOK and it is adjusted to achieve the highest extinction ratio of 14 dB with 50% cross-point of the eye-diagram. The trajectory between the two symbols (on-states and off-states) is straight along the real axis, and therefore the push-pull operation in the silicon MZM provides zero-chirp modulation as well as in the LN MZM.

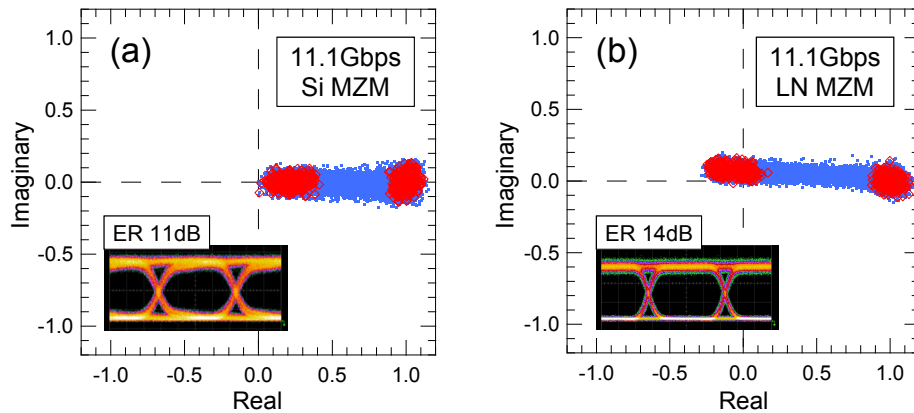


Fig. 2. Constellation diagrams of 11.1-Gbps NRZ-OOK modulation using (a) Si MZM and (b) LN MZM.

The Si MZM was used for the investigation of transmission performance, since zero-chirp operation was confirmed. The measured transmission distance was up to 80-km, which covers a chromatic dispersion range for 2-dB power penalty in BER measurement using the conventional zero-chirp LN MZMs in the 10-Gbps NRZ-OOK format. A transmission range of 60-80 km for 2-dB power penalty at a BER of 10^{-9} [13] and a chromatic dispersion of approximately 1000 ps/nm for 2-dB power penalty at a BER of 10^{-12} [14] were reported using the LN MZMs in the 10-Gbps NRZ-OOK format. Here, in the BER measurements of the Si MZM, different lengths of SMF (20, 40, 60, and 80 km) were used. These SMFs provide positive dispersion parameters. The dispersion parameter of SMF is 17 ps/nm/km at a wavelength of 1550 nm. Further, BER measurements were also performed in negative dispersion parameters using dispersion compensation fiber modules (DCFMs) designed for dispersion compensation of 20-, 40-, 60- and 80-km SMFs.

The modulation format was 11.1-Gbps NRZ-OOK. Pseudo random binary sequence (PRBS) $2^{31}-1$ was applied. The wavelength of the laser was 1545.7 nm. To avoid BER degradation due to fluctuation in clock timing, clock data recovery (CDR) unit was used in the receiver equipment.

The BER characteristics for the Si MZM are plotted in Fig. 3(b) with the waveform after transmission shown in Fig. 3(a). Even in 80-km transmission, the Si MZM has eye-opening and realizes error-free operation without a noise floor. Figure 3(c) shows the dispersion penalty at BER of 10^{-3} for the Si MZM and LN MZM; LN MZM is measured for comparison. Here, the dispersion in the horizontal axis corresponds to the chromatic dispersion in respective fiber lengths for SMFs or DCFMs. The dispersion penalty of the Si MZM is very close to that of the LN MZM, and the dispersion curve is nearly symmetric with respect to the minimum at zero dispersion (back-to-back condition). The zero-chirp modulation in the Si MZM is thus confirmed also in the characteristics of the dispersion penalty. The dispersion tolerance of the Si MZM is more than ± 950 ps/nm at 2-dB power penalty. The slight degradation observed with respect to the LN modulator can be eliminated with further enhancement of the high-speed performance of the Si MZM.

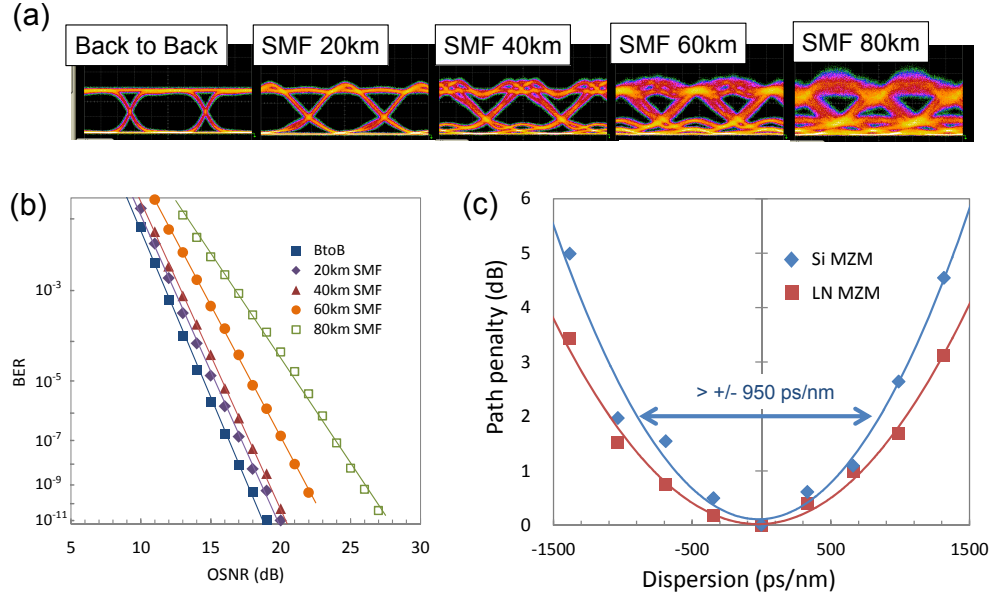


Fig. 3. (a) Eye diagram and (b) BER measurement up to 80-km SMF transmission at 10-Gbps OOK for Si MZM and (c) dispersion penalty at BER 10^{-3} .

4. Frequency chirp of the silicon MZM

Here we discuss the frequency chirp of the Si MZM. The silicon modulator uses plasma dispersion effect for generating phase shift, however it induces the loss change by free carrier absorption in the modulation operation [15]. In push-pull operation, the phase shifts in each arm are opposite each other, and hence, the losses in each arm differ from each other. The resulting imbalance of optical power causes a frequency chirp in the modulation because the cancellation of the phase shift between the two phase shifters is imperfect. Besides the loss change, the linearity between the phase shift and applied bias also affects the frequency chirp. Since the refractive index is decided by carrier distribution change and optical mode, it is not always proportional to the applied bias.

Figure 4(a) shows the relation between the applied bias and the phase shift of the silicon phase shifter. It was obtained from the measurement of asymmetric MZM under DC bias conditions. The relation between the two is linear and similar to that of LN modulator. Figure 4(b) shows the relation between the phase shift and the loss. The obtained loss change is 1 dB per π phase shift.

From these results, we estimated the frequency chirp of the modulator. The electric field of the signal from the MZM is expressed as

$$E = E_i/2 \cdot \exp(j\varphi_c - \alpha_c) [\exp(j\Delta\varphi_A - \Delta\alpha_A + j\varphi_d) + \exp(j\Delta\varphi_B - \Delta\alpha_B)], \quad (1)$$

where E_i denotes the electric field in the input waveguide, φ_c and α_c denotes the phase shift and optical loss common to both arms respectively, $\Delta\varphi_A$ and $\Delta\varphi_B$ denote the phase changes in each arm, $\Delta\alpha_A$ and $\Delta\alpha_B$ denote the loss changes in each arm, φ_d denotes the constant phase difference between the two arms. Assuming that the phase shift and the loss change is proportional to the applied bias and the relation between the two is 1-dB loss change per π phase shift, the loss change is expressed as $\Delta\alpha_A = 1/\pi \cdot (\log_e 10)/20 \cdot \Delta\varphi_A + \alpha_{0A}$, where the α_{0A} denotes constant value. Considering push-pull operation, $\Delta\varphi_B = -\Delta\varphi_A$ under the operation condition of $\Delta\varphi_A$ from $-\pi/4$ to $+\pi/4$. When we choose the initial phase difference $\varphi_d = \pi/2$,

the electric field from the MZM can be calculated as shown in Fig. 4(c). The straight trajectory along the real axis means the signal has modulated without frequency chirp.

For evaluating the practical performance, we simulate the dispersion penalty for the positive and negative dispersion parameters through BER estimation, which is calculated using the simulator OptSim (Synopsys Inc.) Fig. 4(d) shows the simulated dispersion penalties in 11.1-Gbps NRZ-OOK modulation. Here, we simulated three MZMs: (1) an ideal MZM which has no power imbalance nor loss change, and it is considered as a perfect LN modulator, (2) the Si MZM with absorption of 1 dB per π phase shift, and (3) power-imbalanced MZM which has power imbalance corresponding to extinction ratio (ER) of 30 dB. As a result, the dispersion penalty of the Si MZM has shift from the ideal MZM. But the shift is smaller than that of 30-dB ER power-imbalanced MZM. Since typical specification of extinction ratio in LN modulator is 20dB, the 30-dB ER power-imbalanced MZM of can be considered to have sufficiently low frequency chirp characteristics. Thus, the silicon modulator can provide the chirp-free modulation similar to zero-chirp LN modulators.

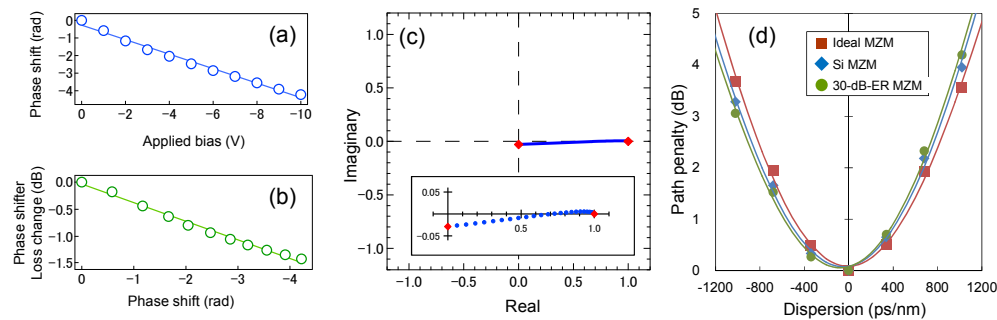


Fig. 4. Relation between (a) applied bias and phase shift, (b) phase shift and loss change, and (c) estimated constellation diagrams, and (d) simulated path penalty for positive and negative dispersion parameters.

The zero-chirp modulation can positively impact PSK modulation because of not only the high tolerance against chromatic dispersion but also because of high tolerance to the imperfections of operation conditions. Figure 5(a) shows constellation diagram of the 22.3-Gbps NRZ-BPSK operation. The driving voltages are $8V_{pp}$ for each arm and the DC bias voltage is $-5V$. Even in 22.3-Gbps operation, zero-chirp modulation with straight trajectory in the symbol transition is observed. In the BER measurement in back-to-back condition, the Si MZM in the 22.3-Gbps NRZ-BPSK format was characterized. For the receiver in this measurement, balanced photodetectors incorporated with 1-bit delay interferometer were utilized in the DPSK direct-detection configuration. The obtained BER characteristics are plotted in Fig. 5(b). The LN modulator was also measured for comparison. The difference between BER for the Si MZM and the LN MZM is less than 0.5 dB at a BER of 10^{-3} . Thus, the Si MZM exhibits BER performance comparable to that of the commercial LN MZM in the 22.3-Gbps NRZ-DPSK format.

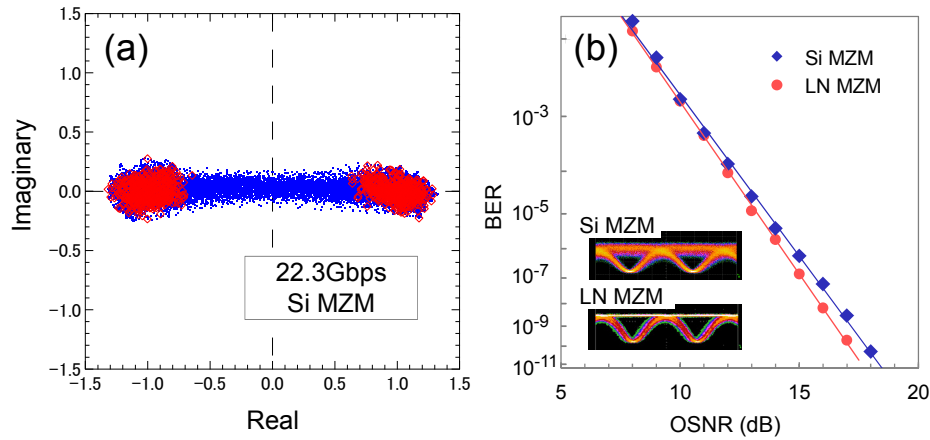


Fig. 5. (a) Constellation diagram of 22.3-Gbps NRZ-BPSK using Si MZM and (b) back-to-back BER of the 22.3-Gbps NRZ-BPSK format.

5. Conclusion

The frequency chirp and transmission performance of the Si MZM have been investigated. We have shown the Si MZM realizes zero-chirp modulation under push-pull operation for 11.1-Gbps NRZ-OOK and 22.3-Gbps NRZ-BPSK in the constellation measurement with homodyne coherent detection. The power penalty of the Si MZM with respect to the commercial LN MZM in the BER measurement is less than 0.5 dB in 22.3-Gbps NRZ-BPSK. Zero-chirp modulation is also confirmed in terms of the symmetric dispersion penalty characteristics by means of BER measurements for transmission up to 80 km using SMF or DCFM.

The zero-chirp and transmission performances of the Si MZM are comparable with those of the LN MZM. Si MZM will provide a low-cost solution as an alternative to LN MZM in high-speed long-haul optical fiber telecommunication systems.